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**PROTECTION OF INTELLECTUAL PROPERTY
RIGHTS FOR PHAMACEUTICAL INVENTIONS IN
VIETNAM IN THE CONTEXT OF INTERNATIONAL
ECONOMIC INTERGRATION**

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SUMMARY OF THE DOCTORAL THESIS

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INTRODUCTION

1. Reason for choosing the topic

Medicines have long been an indispensable weapon in mankind's fight against disease. The use of pharmaceuticals is a necessity in human existence. The Universal Declaration of Human Rights in 1948, adopted and proclaimed by the United Nations General Assembly, affirmed that the right to health protection is one of the fundamental human rights, in which there is an indispensable part is right to access to pharmaceuticals. However, along with the development of science and technology, an equally urgent issue is the protection of intellectual property rights for inventions, including inventions in the pharmaceutical field as IP is also recognized as one of the basic human rights that must be respected. From the perspective of IP law, the owner of a patent has the right to prevent others from using his invention and selling pharmaceutical products containing the invention or drugs produced from the patented process at high prices to recover research and development costs as well as to seek profits. However, from the perspective of patients, especially those suffering from serious diseases that require long-term treatment such as HIV/AIDS, cancer, hepatitis, cardiovascular disease, chronic kidney disease, etc., pharmaceutical prices are too high interfere with their healing process, putting their lives at risk. In addition, paying large sums of money for medicines in the long run can push patients and their families into poverty. The balance between the protection of IP rights for pharmaceutical inventions and the issue of drug access is really a difficult problem for all countries, especially for developing and underdeveloped countries.

As a developing country and suffering from many diseases such as bird flu, HIV/AIDS, tuberculosis, malaria as well as diabetes, cancer... Vietnam is also one of ten most affected climate change countries in the world. In fact, the people's medical needs are very large, but the people's medical expenses are still low and the pharmaceutical costs are too high. The high price of pharmaceutical products is the result of many reasons, one of the main reasons being the establishment of an IPRs protection system for a group of pharmaceutical inventions on a national and international scale. In the national socio-economic development strategy, our Party still emphasizes that national development needs to be accompanied by ensuring social security issues for the people, including health and medical community issues. This point of view of the Party has been thoroughly grasped and concretized in Decision No. 122/QĐ-TT dated January 10, 2013 on the National Strategy to protect, care for and improve people's health for the period 2011 - 2020, with a vision to 2030. In which affirms the view *"... a harmonious combination of socio-economic development with human development in general and human health in particular, is the basis for affirming growth conomic is not only associated with social progress and justice in every step and policy, but also needs to be ensured in each economic development program, plan, scheme, project. Socio-economic development quickly, strongly and sustainably is an*

objective requirement and is only valid when it has a positive impact on people's lives, on the health of the community...".

In addition to the people's requirements for health protection, the Political Report of the 13th National Congress of the Communist Party has identified the task of *"continuing to thoroughly grasp and consistently implement the policy of science and technology as a leading national policy, is a key driving force for the development of modern productive forces, renovating the growth model, and improving productivity, quality, efficiency and competitiveness of the economy. Developing science and technology in line with the general trend of the world and the country's conditions, meet the requirements of national construction and defense in the new period, and adapt to the Fourth Industrial Revolution"*. Strong international economic integration has placed strict requirements for countries to build an effective IPRs protection regime, including products in the field of pharmaceuticals. IPRs protection for products in the field of pharmaceuticals as well as agro-chemicals with their own characteristics and higher protection has always been one of the fiercest debates in the multilateral negotiations IP around the world and so far has not come to an end.

Establishing a strong protection regime for entities in the field of medical and pharmaceuticals is an important step to bring Vietnam deep into the integration cycle. However, that seems to be invisible, pushing the majority of Vietnamese people with low and middle incomes further away from their ability to access scientific and medical advances for disease prevention and treatment, especially in the incurable diseases. On the balance of interests, on the one hand, is the objective requirement of the integration process and the conditions for technology transfer, and on the other hand is the essential need of the majority of people in accessing to meet their health needs, how has Vietnam built a legal system on IPR protection for pharmaceutical inventions, and how has that system been implemented in practice with enforcement measures? Really interested in this issue, the author has chosen the topic to build a thesis reporting the results of PhD program as: *"Protection of intellectual property rights for pharmaceutical inventions in Vietnam in context of international economic integration"*.

2. The object and scope of the study of the doctoral thesis

The object of the research is to focus on researching the provisions of Vietnamese law, the laws of some countries in the world and international treaties on the issue of patent protection for pharmaceuticals inventions, such as the establishment, maintenance and enforcement of intellectual property rights.

With the above research object, the thesis defines the research scope as follows:

- Key theoretical issues related to patent protection for pharmaceuticals inventions such as concepts, characteristics, meaning of protection;

- The provisions of the laws of some countries in the world, important international treaties in the field of intellectual property on the protection of pharmaceutical inventions;
- Current regulations of Vietnamese law on patent protection for pharmaceuticals inventions in comparison with international and national laws;
- Assessing the current situation in granting protection titles and enforcing the protection of intellectual property rights for pharmaceutical inventions in Vietnam at the moment in the context of international economic integration;
- Proposing specific solutions to improve the legal provisions as well as practical activities to improve the effectiveness of intellectual property rights protection for pharmaceutical inventions.

3. Aims and Mission research of the doctoral thesis

* The purpose of the thesis is to clarify theoretical and practical issues of intellectual property rights protection for pharmaceutical inventions in Vietnam today, propose solutions to improve the law and solutions to enhance the effectiveness of patent protection for pharmaceuticals inventions as well as promote the value of this group of subjects in Vietnam in the context of international economic integration.

* To achieve the above purpose, the thesis is built to perform the following main tasks:

- *Firstly*, studying the basic theoretical issues of pharmaceutical inventions as an object of industrial property rights within the scope of IP rights (concepts, characteristics, protection intellectual property rights, distinct from related subjects, etc.).
- *Secondly*, focusing on systematically understanding the provisions of Vietnamese law in comparison with international treaties and the laws of some typical countries on the issue of protection intellectual property rights for pharmaceutical inventions.
- *Finally*, on the basis of understanding theoretical issues, legal provisions as well as practical implementation of protection for pharmaceutical-related inventions in Vietnam, the thesis makes some proposals to improve the provisions of Vietnam's law on this issue in a way that is compatible and ensures international commitments to which Vietnam is a member in relation to ensuring the right to access medicines for Vietnam in the context of strong integration.

4. Research Methodology

The thesis is researched on the basis of the methodology of Marxism - Leninism, Ho Chi Minh thought; Besides, it is applying the guiding viewpoints, lines and policies of the Party and the State's laws on the development of the IP field in the context of integration.

On the basis of the above-mentioned methodology, in the research process, the thesis uses specific research methods such as: analysis, synthesis, comparison combined with statistical modeling methods. Specifically:

- Traditional methods such as analysis, synthesis and comparison are commonly used when studying the provisions of international treaties, the laws of other countries and the practical laws of Vietnam; In particular, the thesis focuses on comparative jurisprudence to consider and evaluate the provisions of Vietnamese law in relation to national laws and international law in the context of integration.

- In addition, statistical modeling methods are used effectively and appropriately to clarify the issues that the topic researches and presents such as the statistics of the situation registration and processing of pharmaceuticals patent applications; modeling the application processing process as well as highlighting the unique features of the application processing process of pharmaceutical inventions; summarizing data on the infringement of rights of invention owners; comparison tables...

5. Scientific significance and novelty of the thesis

- The thesis is the first work to comprehensively approach the key theoretical issues related to the protection of intellectual property rights for pharmaceutical inventions, specifically: (i) clarifying the conceptual content of pharmaceutical inventions and their characteristics; (ii) distinguish the subject matter of a pharmaceutical invention from other subjects also in the pharmaceutical field; (iii) develop a scientific concept of intellectual property rights protection for pharmaceutical inventions; (iv) learning, analyzing and making judgments about the characteristics, advantages and disadvantages in the laws of countries around the world in the matter of patent protection related to pharmaceuticals; (v) analyzing and connecting the provisions of international law on patent protection related to pharmaceuticals in the context of strong integration...

- The thesis is an independent and systematic study of the provisions of Vietnamese law in various documents regarding the protection of intellectual property rights for pharmaceutical inventions. Through the concentrated and in-depth analysis of the practical law on this issue, the thesis points out the shortcomings of the Vietnamese law on patent protection for pharmaceutical inventions.

- The thesis collects information and data to make general judgments and assessments about the current status of intellectual property rights protection for pharmaceutical inventions in Vietnam from the rights establishment as well as rights protection for the owners.

- On the basis of concentrated and comprehensive studies on theoretical and practical issues, the thesis offers specific and practical solutions to overcome existing limitations in

the issue of intellectual property rights protection for pharmaceutical inventions in Vietnam today.

6. The practical significance of the doctoral thesis

The research results of the thesis are valuable documents for teaching and researching jurisprudence in the field of intellectual property for training institutions. In addition, the results of the project are also documents for professional agencies to refer to when developing and perfecting the law on intellectual property, specifically for objects of pharmaceutical inventions; Officials and specialized agencies related to the field of pharmacy can use it as a document to improve their knowledge and draw lessons for themselves in actual work.

7. Structure of the doctoral thesis

The thesis includes introduction, content, conclusion, list of references, appendices. The content is arranged into 04 (four) chapters, with sub-conclusions of each chapter, including:

- ❖ *Chapter 1: Overview of the research situation in other countries and in Vietnam related to the thesis topic*

- ❖ *Chapter 2: Basic theoretical issues on protection of intellectual property rights for pharmaceutical inventions*

- ❖ *Chapter 3: International law and the laws of some countries on the world on the protection of intellectual property rights for pharmaceutical inventions*

- ❖ *Chapter 4: Law and practice in Vietnam on protection of intellectual property rights for pharmaceutical inventions and some proposals and solutions to improve related laws in the context of international economic integration*

CHAPTER 1

OVERVIEW OF THE RESEARCH SITUATION IN OTHER COUNTRIES AND IN VIETNAM RELATED TO THE THESIS TOPIC

Through research, it can be seen that the number of studies on this issue is abundant, contributing to building a panorama of international law and national law on the protection of intellectual property rights for pharmaceutical inventions. However, as pointed out, the research works that have been done not yet fully and systematically covered the theoretical issues related to the implementation of protection of intellectual property rights for pharmaceutical inventions as well as the limited exercise of the rights of the owner stem from the nature of this object greatly affecting the public interest. Most of the works stop at the presentation of a mechanism and solution for how to balance the different interest groups related to the object of invention in the pharmaceutical field without exporting from any theoretical basis. In addition, there are almost no studies on the practice of protecting

and enforcing intellectual property rights for this group of subjects, both at home and abroad.

On the basis of comparison with the purpose, research tasks of the thesis and limitations of the done works, the thesis will clarify the following issues:

In theory, the thesis will be the study and systematic presentation of theoretical issues related to pharmaceutical patent protection such as defining the connotation of the pharmaceutical inventions; characteristics and theoretical basis of the protection of intellectual property rights for pharmaceutical inventions; the relationship between intellectual property rights for pharmaceutical inventions and people's access to medicines to protect health; Some specific legal issues related such as the issue of expiration of rights, the issue of compulsory transfer of the right to use inventions, etc.

In terms of legality, the thesis will analyze and comment on the provisions of international treaties and the laws of some countries on the protection of pharmaceutical inventions, especially the member countries of the TRIPS Agreement in the implementation of provisions of this Agreement for the protection of pharmaceutical inventions. In order to create logic in the presentation and facilitate comparison between international law and the laws of countries, the writer will divide the content according to legal issues of intellectual property rights protection for pharmaceutical inventions from the establishment of rights to the enforcement of rights.

Finally, regarding the legal and practical issues of Vietnam, the thesis conducts a comprehensive analysis of the provisions of Vietnamese law on the protection of pharmaceutical inventions, the current status of IPR protection for this group. On the basis of presenting the inadequacies and limitations of the law as well as the practice of protection, the writer will go into the explanation of the causes of the limitation; at the same time clearly define the requirements in perfecting the law and improving the capacity to enforce IPR protection for pharmaceutical inventions in Vietnam; thereby proposing specific and reasonable solutions to meet the urgent needs in the enforcement of IPR protection in Vietnam in the context of economic integration as well as ensuring the interests of the people in accessing essential medicines to protect health. In the process of analyzing these contents, by comparing and contrasting methods, the writer will integrate the experiences drawn from the study of the law and practice of IPR protection for pharmaceutical inventions of other countries to highlight the recommendations for Vietnamese legislation.

CHAPTER 2

BASIC THEORETICAL ISSUES ON PROTECTION OF INTELLECTUAL PROPERTY RIGHTS FOR PHAMACEUTICAL INVENTIONS

2.1. Basic concepts and features of pharmaceutical inventions

2.1.1. Concepts of pharmaceutical inventions

From researching, comparing contents and provisions in international treaties, national laws, domestic legal documents, the author draws out the concept of pharmaceutical inventions as following: *pharmaceutical inventions are patentable objects including products, processes, methods or any other object used to solve problems in the field of medicine and public health, specifically for the prevention, diagnosis and treatment of disease in humans or animals, by the application of natural rules.*

At the patent offices of Vietnam as well as of countries around the world, many other forms can be considered to be granted a patent for SCLQDDP, however, they can be divided into two main groups: products and procedure.

2.1.2. Essential features of pharmaceutical inventions

Like other objects of industrial property rights, pharmaceutical-related objects, if they want to be protected in the name of invention, must be capable of industrial application in the fields of science and technology in order to create produce valuable products for human life as well as productive labor activities. First of all, pharmaceutical inventions have all the same characteristics as other objects of industrial property rights. Next, pharmaceutical inventions must meet the same standards as any invention in other fields, having all the characteristics of a common invention.

Besides the general characteristics, it can be seen that there are characteristics that only subjects related to pharmaceuticals or in other words, subjects in the field of medical pharmaceuticals have. These characteristics have led to an inevitable consequence that there are many very specific legal provisions for this type of object.

Firstly, pharmaceutical inventions are those aimed at solving public health problems. These objects have the effect or contain ingredients that have the effect of treating or preventing diseases in humans or animals, or are used for the purpose of restoring, adjusting, changing physiological functions, or used for medical diagnostic purposes in humans or animals.

Secondly, pharmaceutical inventions are not like most inventions in other technical fields, it has to go through a research and manufacturing process with very strict principles. Usually, a new pharmaceutical product that we often use as a drug for short, is created by a process that includes the basic steps of selective search, animal experiments, experiments on humans, apply for a patent, apply for a license to circulate drugs.

Thirdly, pharmaceutical inventions are subject to great influence on the interests, national security and public health, so the patent law is associated with regulations on the scope and limitations of the rights of the owner by monopoly. Many countries around the world,

especially developing countries, have expressed their great concern that the implementation of strong IP regimes could affect efforts to improve public health and patents. Drug and treatment restrictions can impede governments' efforts to address urgent policy problems with access to affordable health care, which in turn also complicates public health programs.

Fourth, pharmaceutical inventions has a shorter actual protection term than other objects of intellectual property rights.

2.1.3. Typical types of claims for pharmaceutical inventions

Researched and created in the field of public health, subjects are considered for patent protection in various forms. Some common forms are commonly considered under national legislation such as: Claims in the form of preparations and ingredients, Claims in combination, Claims for dosage/amount, Claims salt, ether and ester claims, Treatment claims, Usage claims...

Depending on the level of economic development and views on the establishment of IPR protection for pharmaceutical inventions, countries around the world choose whether to accept or not accept protection for various forms this subject.

2.1.4. Protection of confidential information for scientific test data of pharmaceutical inventions

For products in the pharmaceutical field, a special requirement is set that to be licensed for circulation and use, inventors as well as businesses must provide state agencies authoritative test data for such pharmaceutical products to demonstrate safety and efficacy. This data will be the basis for the relevant authority to decide whether to proceed with the licensing of this new pharmaceutical entity or not. Test data is important not only for health purposes but also for the environment. In addition, data is important for commercial purposes as data availability is a condition for obtaining marketing approval for new products, modification or new use of existing products. have.

In the pharmaceutical sector, most international and national regulatory instruments have requirements to varying degrees with regard to the confidentiality of test and other data.

2.2. Protection on intellectual property rights for pharmaceutical inventions

2.2.1. Concept of protection intellectual property rights for pharmaceutical inventions

Protection of pharmaceutical inventions means that the State establishes and recognizes the rights of a subject to pharmaceutical inventions through the protection title of a patent, maintains and ensures the protection of pharmaceutical inventions through the provisions of law, maintain the rights of holders within the limits and scope of the granted protection title and to ensure that the rights recognized in such protection title are exercised in practice.

In the framework of the thesis, the concept of "*protection*" is interpreted in a very broad sense, not only related to the establishment of rights, maintaining the validity of protected rights but also measures to ensure the owner's rights are actually implemented.

2.2.2. Rationale for the protection of intellectual property rights for pharmaceutical inventions

The theoretical bases for the establishment of the IPR protection regime for the pharmaceutical inventions are specifically explained as follows:

(i) *Firstly*, pharmaceutical inventions are a legal (intellectual property) and valuable property of the owner. The right to pharmaceutical inventions is recognized and protected by international and national law as a legal right of the owner of this intellectual property. Through the process of reviewing and granting protection titles prescribed by the State, the State grants the owners rights similar to ownership rights over other types of property, naturally associated with the specific representation of intellectual property rights. The owner's right to the pharmaceutical inventions is a type of right arising on the basis of registration and recognized and protected by a competent state agency.

(ii) *Secondly*, Protection of pharmaceutical inventions is the basis for the development of research, creation and promotion activities of science and technology for the advancement of mankind. The development of the pharmaceutical industry depends heavily on the IP system and patents are considered as a tool to encourage the research and development of new drugs. Patent protection encourages the disclosure of new technologies and the transfer and dissemination of technologies in general, and in the pharmaceutical sector particularly.

(iii) *Thirdly*, Protection of pharmaceutical inventions creates and develops healthy competition in international trade and economic exchange. The protection of IP rights in general helps prevent misleading acts in commercial activities, ensuring fair competition.

(iv) *Fourth*, for the country, Protection of pharmaceutical inventions will help encourage and promote investment development and technology transfer to the country. For the country, when fully implementing international commitments on the protection of IP rights, including the issue of protection of pharmaceutical inventions, it will help encourage and promote investment development and technology transfer for the country. Through this path, countries, especially developing and underdeveloped countries, will have the opportunity to quickly access advanced technology to carry out industrialization and modernization of the country. Fully and effectively implementing international commitments in the field of IPR protection for inventions is also the key for countries to integrate into the world economy, especially in the current period, when IP becomes an important content of international trade relations.

(v) *Fifth*, the establishment of patent protection for pharmaceutical inventions should impose certain limitations due to its dual nature and negative effects. Especially for developing countries and less developed countries, protection of pharmaceutical inventions brings more negative effects than positive. The negative effects that the patent system can bring include: influence on competition, influence on product selling price, effect on society's ability to access drugs, influence to the development of the pharmaceutical industry.

2.3. Legal basis for conducting protection intellectual property rights for pharmaceutical inventions

2.3.1. International law on the protection of intellectual property rights for pharmaceutical inventions

In the early stages, patent law was limited to a few bilateral treaties. This remained until a conference was held in Paris in 1878 and ratified the Paris Convention of 1883 for the protection of industrial property rights. Then, to meet the needs of economic integration and development among countries around the world, many international treaties at different levels (international, regional, bilateral) were born. Important and typical treaties can be mentioned such as the TRIPS Agreement, the CPTPP Agreement, the EVFTA Agreement, etc. In the content of these treaties, there are always specific provisions related to pharmaceutical inventions. From the great distance between the views of developed and developing countries, gradually countries have found a common voice on the establishment protection of IPRs for pharmaceutical inventions.

2.3.2. National law on the protection of intellectual property rights for pharmaceutical inventions

Even in developed countries, where intellectual property rights in general and inventions in particular have been protected for hundreds of years, patent protection for pharmaceuticals has only just begun in the last years of the twentieth century; eg France since 1960, Switzerland since 1977, Japan since 1976, Italy, Sweden since 1978, Spain, Portugal, Greece and Norway since 1992... The usual reason is given proposed to deny patent protection in the pharmaceutical sector is motivated by "the social importance of pharmaceuticals as well as the belief that the protection of pharmaceutical inventions would lead to the exclusive abuse of the owner", which greatly affects the general welfare of society. It is in the countries that recognize the patenting regime for pharmaceutical inventions that also stipulates a lot of restrictions on the rights of the owner during the term of protection.

Throughout the history of formation and development of the IP legal system in the world, regarding the establishment of the protection regime for intellectual property rights, there exist two main opposing views: (i) The first point of view belongs to developed countries, capitalist countries in which the US is the leading country. These countries consider human intellectual

property to be sacred and inviolable, so protection needs to be established to the highest degree, laws need to be formulated and implemented most effectively in protecting the rights of owners, which creates a high motivation for the development of intellectual property and science and technology; (ii) The second point of view belongs to the group of developing and least developed countries, including Vietnam. According to the arguments of these countries, intellectual property is the product of creative labor, it is a product of the private property of each individual, but ultimately also of humanity in general, the protection should be established in a mode suitable to the economic, political and social conditions of each country and the needs of the majority of the people. The law should therefore achieve a level of flexibility with respect to the protection of the rights of the owner.

The two viewpoints mentioned above are even more evident and sharper when it comes to the protection of IPRs for pharmaceutical inventions.

2.3.3. Interaction between international law and national law in the protection of intellectual property rights for pharmaceutical inventions

The protection of IPRs in general and IPRs for pharmaceutical inventions in particular was originally implemented on the basis of the national legal document system. However, due to the limited territorial specificity of the IPR legislation of countries, the loss of protection of objects in countries other than the original country in which the right of the subject is derived to the requirement to establish international protection methods. The treaties have established minimum standards for providing protection to the subjects of IPR including pharmaceutical inventions. On that basis, the member countries will develop in their legal system specific contents. On the one hand, the laws of other countries will have to comply with the frameworks established by the treaties, and on the other hand, will make the most of the gaps and flexibility of the treaties to suit the conditions development of their own country, bringing into full play the value of the IPRs protection system.

CHAPTER 3

INTERNATIONAL LAW AND THE LAW OF SOME COUNTRIES ON THE WORLD ON THE PROTECTION OF INTELLECTUAL PROPERTY RIGHTS FOR PHARMACEUTICAL INVENTIONS

3.1. International legal principles for the protection of intellectual property rights for pharmaceutical-related inventions

The two legal regimes that are also considered as two basic principles in the protection of IPRs in international law are recognized by treaties as The National Treatment (NT) and the The Most Favored Nation (MFN). For the protection of inventions in the pharmaceutical sector, countries must always comply with the minimum principles of protection as for any other object of IP rights.

3.2. Scope, criteria and term of protection for pharmaceutical inventions

3.2.1. Scope of protection

From the recognition of the meaning as well as the effects of the establishment of the IPR protection regime for pharmaceutical inventions, international and national laws always have limitations on the scope of protected objects in this field. Common restrictions include the exclusion from patentability of inventions that need to be prohibited from commercial exploitation in its territory in order to protect human and animal life and health or plant food, to avoid causing serious harm to the environment; refused to grant patents for substances found in nature...

3.2.2. Criteria for protection

According to the provisions of international law and national law, an object that satisfies and meets the following criteria will be granted a patent:

- Novelty: interpreted and applied in different ways depending on the laws of the country as well as the interpretation of the patent authorities and courts in that country, especially in the case of define the limits of the circumstances and scope of what has been previously disclosed so that an object is not considered to have lost its novelty.

- Level of innovation: countries also choose the level of "unclear" or "non obvious" for people with average qualifications in the pharmaceutical sector to determine whether the level of innovation is achieved.

- Industrial applicability: this requirement is considered and selected between the levels of industrial applicability or utility of the object depending on the point of view of protection and the level of progress in industry national pharmaceuticals.

3.2.3. Term of protection

The term of protection for a patent application is basically provided for under international treaties of 20 years from the date of filing of the application for registration, however, depending on the point of view of each country, there are provisions to accept the extension of protection time or not.

3.3. Rights and obligations of patent holders pharmaceutical inventions

3.3.1. Establishing intellectual property rights for pharmaceutical inventions

The establishment of intellectual property rights for pharmaceutical inventions mainly revolves around the main contents, which are the process of filing a claim for protection and examining the application for protection and granting a protection title; the disclosure of information in the application for protection.

(i) Regarding the procedure for filing, examining and granting a protection title: most of the treaties do not state it, but rather will be provided for by national legislation. Most inventions go through a “substantive review process” in which the patent application is scrutinized by a patent examiner, comparing the relevant technical status to the application

with the claim stated in the application. Some countries still maintain a “registration system” whereby an applicant receives a patent after completing the necessary formalities without examining the content of the patent application. In this system, it is not until the patent is sued in court that it is determined whether the invention is invalid or not because only then can the technical situation be studied and evaluated in terms of content.

(ii) Regarding the provisions on disclosure of information in the application for protection: The laws of most countries and international treaties stipulate that the grant of a patent is often conditional on full disclosure sufficient for the invention, the specifications must provide information to enable a person skilled in the art field to create or practice the requested invention.

In addition, for the pharmaceutical inventions, the regulation on drug marketing licensing procedures is a separate procedure specific to this group of subjects, which is also recognized in most international treaties and national laws. different levels and requirements.

3.3.2. Rights and obligations of patent owners related to pharmaceutical inventions

The international treaties create space for national legislation to specify the rights and obligations of the owner of the pharmaceutical inventions. The exercise of intellectual property rights by the right holder within the scope and term of protection as prescribed by law must not infringe upon the interests of the State, public interests, and legitimate rights and interests of organizations and individuals, must not violate other provisions of relevant laws. In the case of ensuring national defense, security, people's livelihood and other interests of the State and society, the State has the right to forbid or restrict the IPR holder from exercising his/her rights or force the right holder to IP must allow other organizations and individuals to use one or more of its rights under suitable conditions.

3.3.3. Restricted rights of patent owners related to pharmaceutical inventions

IPR rights to an pharmaceutical inventions give the owner the right to prevent others from using and exploiting a protected IPR. However, because of the great influence on the interests of the community, the restrictions placed on the rights of the owner of industrial property objects in general are mainly applied to this type of object. Both treaties and national laws provide for some exceptions where someone else can use or exploit a protected patent without the consent of the patent owner under the licensing contract of the exclusive right to use the invention without being considered a violation of the law; the issue of compulsory licensing of patents; regulations on the issue of expiration of rights and parallel import of pharmaceuticals.

3.4. Enforcement of intellectual property rights for pharmaceutical inventions

3.4.1. General comment

Proposing a system of measures to enforce the right to protection for pharmaceutical inventions is an important content to ensure that the rights of owners are promoted in practice.

3.4.2. Measures to enforce intellectual property rights for pharmaceutical inventions

Internationally, the TRIPS Agreement is the most advanced and complete international treaty in establishing a system of measures to enforce rights such as civil measures, criminal measures, administrative measures, and control border measure. The TRIPS Agreement also stipulates that countries are not required to establish a judicial system for the enforcement of IPRs separate from the judicial system for law enforcement. In addition to providing for adequate measures that members must take to protect their patent rights, the TRIPS Agreement also has certain limitations to ensure balance with the interests of society. Specifically, in the procedures for enforcement of rights, the TRIPS Agreement clearly shows the view that the rights to inventions, including pharmaceutical inventions, are a civil right and civil procedures will be applied to patent infringement acts.

National legislation on the basis of the requirements of the TRIPS Agreement chooses to develop specific contents and ways to take measures to enforce IPR rights to pharmaceutical inventions.

CHAPTER 4

LAW AND PRACTICE IN VIETNAM ON PROTECTION OF INTELLECTUAL RIGHTS FOR PHAMACEUTICAL INVENTIONS AND SOME PROPOSALS AND SOLUTIONS TO IMPROVE RELATED LAWS IN THE CONTEXT OF INTERNATIONAL ECONOMIC INTERGRATION

4.1. Overview of the history of formation, development and international integration of Vietnamese law on protection of intellectual property rights for pharmaceutical inventions

In Vietnam, pharmaceutical products have been protected as patents since the birth of this protection system. In the early years, protection for patents was generally different from that of most other countries in the world. Along with each step of integration, the patent protection mechanism related to pharmaceuticals has been gradually changed and so far that it has basically met international standards.

4.1.1 The period of centrally planned economic development from 1981 to 1986

This period marked an important milestone with the birth of the Industrial Property Ordinance as the basic legal foundation for the protection of IP rights for industrial property objects, including the pharmaceutical inventions. Thanks to that, the patent protection mechanism in the pharmaceutical field has had a fundamental change in principle,

completely abolishing the form of granting patents, and pharmaceutical inventions like objects in other fields is left the only form of protection is the grant of a patent, a patent with all the same features as the protection of inventions internationally and in other countries.

The civil aspects of industrial property relations as well as industrial property rights are shown in a very faint manner, on the contrary, the administrative nature of industrial property relations is emphasized, especially, the infringement of industrial property rights is considered a violation of the law, disputes over industrial property are considered administrative violations and administrative measures are considered as the main tools to handle disputes and infringements.

4.1.2. The period of renewal and initial international integration began in 1987 with the birth of the Civil Code 1995

The Civil Code 1995 was born, becoming the highest legal basis for the comprehensive implementation of IPR protection activities for Vietnamese inventions, including the pharmaceutical inventions. The system of IPR protection in general, including pharmaceutical inventions in particular, has been raised to the highest level on the basis of documents promulgated by the legislature, not just regulated by sub-law documents as before. Industrial property rights are affirmed as a civil right protected by law like other property rights. Although it has been enhanced in terms of legal validity, the IPR protection system still basically contains the same limitations as the previous period. This is because in terms of content, Part VI of the 1995 Civil Code almost keeps the corresponding provisions of the Industrial Property Ordinance, in which most of the provisions are for the definition and determination of the content/scope as well as the basis for the arising of rights.

4.1.3. The period of full, comprehensive and strong international integration since the 2005 Intellectual Property Law until now

Stemming from the requirements in the process of negotiating to join the WTO as well as objective needs, Vietnam has made many efforts in amending the IP law in order to meet the requirements of IP protection according to the standards. the minimum requirements of the TRIPS Agreement, but at the same time, it is still possible to make good use of the exceptions provided for by the TRIPS Agreement to balance the interests between IPR owners and the common interests of the whole society. Therefore, the 2005 IP Law was more complete and superior and many guiding documents were issued to replace the part of the IP law in the 1995 Civil Code. The birth of the 2005 IP Law in the context of integration with requirements for the implementation of international commitments to which Vietnam participates, with many guiding documents, has created a relatively complete and complete legal system to protect IP rights for IP objects, including inventions, and pharmaceutical inventions is also a protected area.

4.2. Current status of Vietnamese law on protection of intellectual property rights for pharmaceutical inventions

4.2.1. Protection criterias, scope and duration of protection for pharmaceutical inventions

(i) Protection criterias: pharmaceutical inventions has always been the subject of protection in Vietnam. Based on Article 59 of the Intellectual Property Law, pharmaceutical products and pharmaceutical manufacturing processes are protected in Vietnam because they are not on the list of excluded objects that are not protected in the name of inventions. On the basis of the minimum standards set forth by the TRIPS Agreement, Vietnamese law also stipulates the conditions for a protected area of land use rights. Accordingly, the conditions to be protected in Vietnam for patents are similar to those of inventions in other fields. Specifically, a patent will be protected in the form of a patent if it meets the conditions of novelty, inventiveness and industrial applicability.

(ii) Scope of protection: the use of a known substance for a new function is not considered the subject of protection, the law does not protect the invention in the form of use (use claim); a number of pharmaceutical-related objects are also excluded from protection in the name of an invention that derives from the character of each class or a social purpose; Products that are natural, not man-made, will not be considered technical solutions and are not necessarily patentable.

(iii) Term of protection: the term of a protection title for an invention is 20 years from the date of issue and is valid for 20 years from the filing date. Besides, it can be protected as a utility solution with a term of protection of 10 years from the date of filing. This type of term is defined and not renewable.

4.2.2. Establishing intellectual property rights for pharmaceutical inventions

Current Vietnamese law stipulates the process of appraising a patent application, which goes through all the steps of formal examination and substantive examination.

In addition to ensuring the general requirements for applications of other fields, the application for registration of pharmaceutical inventions also has specific regulations and requirements. In particular, the patent description must state the results of clinical trials and the pharmacological effects of the pharmaceutical product, including at least the following information: substance/mixture used; test method (system) used; test results; correlation between the results of pharmacological effects obtained in the trial with the actual application of pharmaceutical products in the prevention, diagnosis and treatment of diseases.

4.2.3. Rights and obligations of patent owners related to pharmaceuticals inventions

According to the provisions of Vietnamese law, holders of pharmaceutical patents as well as inventions in other fields have the following basic rights: (i) owners of patents granted protection titles have the exclusive right to use these patents. use, permit others to use their inventions; (ii) the owner of the property has the right to prevent others from using

the object under his/her ownership; (iii) the owner of the SCLQDDP patent has the right to dispose of the object under his/her ownership in many ways.

In addition to the basic rights prescribed for owners of industrial property objects on the basis of the provisions of the TRIPS Agreement, to ensure that the owners can better exercise the contents of their rights to the Industrial property subject, Vietnamese law also recognizes temporary rights to owners of industrial property objects.

Obligations that the owner must perform: (i) the owner is obliged to pay remuneration to the creator of the invention on the basis of mutual agreement between the parties in accordance with the provisions of law; (ii) the owner has an obligation to use the protected invention, in particular an obligation to manufacture the protected product or apply a protected process to meet the needs of national defense, security, prevention disease, treatment, nutrition for the people or other urgent needs of the society; (iii) the obligation to authorize the use of the underlying invention in order to use the dependent invention.

The law, in addition to stipulating rights and obligations for the owner of a patent for an invention, also provides for restrictions on the rights of this subject. These are cases where the patent owner is not entitled to prevent others from using his invention; compulsory licensing of inventions and use of inventions in the name of the State; exhaustion and parallel import.

4.2.4. Enforcement of intellectual property rights for pharmaceutical inventions

Vietnam's IP law does not stipulate IPR infringement acts for each specific object of industrial property, but infringements of rights to inventions including patents will be determined based on common signs (acts of infringing upon rights to novel industrial property objects include inventions, industrial designs, and layout designs of integrated circuits). It can be determined that acts of infringing upon IPRs for pharmaceutical inventions are acts performed that infringe upon the patent when this invention is in the protection period. Accordingly, acts that are considered to be infringing IP rights for patents include the following two basic types: (i) Using pharmaceutical inventions within the validity period of the protection title without permission of owner; (ii) Using pharmaceutical inventions without paying compensation under the provision of provisional rights to the invention.

On the basis of determining the acts that are considered to be infringing on IP rights for pharmaceutical invention, the law of Vietnam has developed a system of enforcement measures to protect the rights of the owner of the civil law, against acts of infringement. infringement by other subjects while the invention is in the term of protection.

4.3. The practice of protecting intellectual property rights in Vietnam for pharmaceutical inventions

4.3.1. Status of registration and grant of protection titles for pharmaceutical inventions in Vietnam

From the research and statistics obtained, some key points can be clearly seen in the situation of registration and grant of protection titles for pharmaceutical inventions:

(i) Firstly, the pharmaceutical inventions was only really registered since 1990, that is, after the Ordinance on the Protection of Industrial Property Rights was promulgated.

(ii) Secondly, since the promulgation of the Law on Intellectual Property in 2005, it can be seen as a radical change to create a solid legal basis for the protection of IPR pharmaceutical inventions in Vietnam. Therefore, statistics show a sudden increase in the number of applications and the number of patents granted at this stage.

(iii) Thirdly, along with the growth in the number of patent applications in general, pharmaceutical patent applications always account for a large proportion of the total number of patent applications in all technical fields.

(iv) Another fact is that most of the pharmaceutical inventions registered in Vietnam are owned by foreign entities, the number of patents of Vietnamese owners is negligible.

4.3.2. The situation of law enforcement on the protection of intellectual property rights for pharmaceutical inventions

The situation of law enforcement on the protection of IPRs for pharmaceutical inventions can be drawn on the following key points:

(i) Firstly, patented inventions have been put into use by their owners in Vietnam, and have exercised the content of their rights to use within their rights recognized by law. In addition, the owner has also exercised the right to prevent others from using his invention and take preventive measures such as warning, negotiating, complaining, and suing the infringer on a case-by-case basis. However, from the meager data on patent-related disputes in this field, it can be seen that the majority of patents granted in Vietnam are either unused or not. is infringed or has been infringed but the owner is not interested in fixing it, there is no request to the competent authorities to resolve...

(ii) Secondly, on the part of the enforcement agencies, although the number of agencies with the function of ensuring law enforcement in this regard is very large. However, the data shows that so far, only very few complaints/complaints about IP infringement have been recorded (none of which are related to pharmaceuticals).

(iii) Thirdly, on the part of specialized agencies: although they do not have the function of handling disputes and infringement of IP rights, the two agencies, the NOIP and the Institute of IP Science, play an important role in their activities this and the fact is to provide more complete, truthful information and data, and more accurately reflect the situation of infringement of rights to pharmaceutical inventions.

4.4. Some proposals and recommendations to improve the law on protection of intellectual property rights for pharmaceutical inventions in Vietnam in the context of international economic integration

4.4.1. General direction

The improvement of the provisions of Vietnamese law related to the protection of IP rights for pharmaceutical inventions needs to ensure compliance with the general guidelines of principle:

(i) Completing the law on protection of IPRs for pharmaceutical inventions must be consistent with the requirements of perfecting the socialist-oriented market economy institution, building a socialist rule of law state in Vietnam of the people, by the people, for the people.

(ii) Completing the law on protection of IPRs for pharmaceutical inventions in accordance with international practices, complying with international commitments to which Vietnam is a member, ensuring international integration, and at the same time researching and exploiting effective provisions on the exceptions and limitations of IPRs provided for in treaties.

(iii) Ensure the principle of balancing the interests of patent holders of pharmaceutical patents with the common interests of society.

4.4.2. Some solutions for developing and completing legal documents on intellectual property rights protection for pharmaceutical-related inventions

As mentioned, as a country ranked in the group of developing countries in the world, Vietnam, on the one hand, needs to comply with its international commitments, build a regime to protect intellectual property rights for pharmaceutical inventions with minimum protection requirements and conditions. On the other hand, we clearly need to take advantage of the flexibility that international agreements create to be able to promote and take advantage of available resources. The solutions that the author proposes to improve the law on protection of IP rights for civil wars are divided into groups of research issues:

i) Proposing related to regulations on conditions and scope of patent protection pharmaceutical inventions:

- Should expand and create a stronger and clearer legal basis for the target group of traditional medicine drugs within the scope of subjects that need to be protected for IPR.

- It is necessary to develop specific regulations on restricting the provision of patents for drugs related to infectious diseases, serious diseases and having a large impact in the community.

- Develop regulations on compensating for delays caused by drug marketing authorization under the provisions of the EVFTA Agreement.

(ii) Proposing deals with provisions on the scope of rights and obligations of patent owners and other related subjects.

- Proposal to amend regulations on the obligation to use inventions of owners.

- Add the limited cases of exceptions prohibiting others from using the invention.

- Amend the regulations on the right to use pharmaceutical drugs granted compulsory licenses, in addition to meeting the goal of supplying to the domestic market first, but also

can be supplied to the market in countries that do not have sufficient conditions to produce drug under a compulsory license decision of the host country that is a member of the WTO.

- Concretize and finalize the regulations on compulsory licensing to take advantage of this provision on the basis of conformity with the standards and requirements of the TRIPS Agreement.

- Make specific regulations to serve as a basis for identifying and sanctioning cases of improper use of IP rights by the owner.

4.4.3. Law enforcement solutions to protect intellectual property rights for pharmaceutical inventions and other solutions

On the basis of perfecting the legal provisions on the protection of IP rights for the civil cod pharmaceutical inventions, it is necessary to carry out synchronously many solutions for the protection of rights to be effective in practice. Some solutions suggested by the author:

- (i) Invest in training human resources with deep expertise in the field of pharmaceuticals in agencies related to the appraisal and grant of protection titles for pharmaceutical inventions as well as law enforcement IPRs agencies on protection of intellectual property rights.

- (ii) Developing the domestic pharmaceutical industry, improving the pharmaceutical production and research and development capacity of domestic enterprises; at the same time, raise the awareness of enterprises about IP law in general, and about protection of civil property in particular.

- (iii) Establishing an international cooperation network in the field of research, development and protection of intellectual property rights for pharmaceuticals.

GENERAL CONCLUSION

1. The history of formation and development of international law on the protection of intellectual property rights has always had many sharp contradictions and conflicts among many related subjects. One of the most stressful issues raised throughout all negotiations is the level of protection for pharmaceutical inventions. Stemming from the importance of this object to the public issue of human health and life, the establishment of a protection regime for it has been faced with many different obstacles opinions. On the one hand, rights holders argue that they need a fair compensation for the effort and intelligence expended in order to be able to continue to reinvest and thus encourage creativity and development and innovation technology. On the other hand, the majority of people and those on the side of the public interest think that the establishment of exclusive protection for this group of people will make it more difficult to access pharmaceuticals, difficult to solve health security problems at the national and international level.

Concluding these controversies, the Paris Convention and the TRIPS Agreement are important multilateral international treaties in the field of IPRs protection that have been

established and require member states to provide exclusive protection for inventions in all fields of science and technology, so inventions related to pharmaceuticals are no exception. However, the TRIPS Agreement still gives member countries flexibility and the right to take the initiative in establishing a system of patent protection in order to achieve the most important goal set out by the Agreement. In addition, to emphasize the points of flexibility for member states in dealing with public health issues, the Doha Declaration on TRIPS Agreement and public health was born.

2. Under the requirements of international treaties on the protection of industrial property rights, the most important of which is the TRIPS Agreement, the law of Vietnam has built a system of IPRs protection for pharmaceutical inventions. Documents referring to the protection of this group of subjects are listed from the Intellectual Property Law, the Pharmacy Law, the Civil Code, the Penal Code, the Ordinance on Sanctions of Administrative Violations, and guiding documents. guide the implementation of the above laws. With the legal documents, the provisions on IPRs protection for pharmaceutical inventions have been relatively comprehensive and complete from the establishment of rights, content of rights and limitations of rights, enforcement measures. The law of Vietnam ensures compliance and compatibility with international treaties to which it is a member, on the other hand, it is also flexible and sensitive to the actual conditions of the country.

3. The current status of registration and grant of protection titles for pharmaceutical inventions in Vietnam shows that there are many shortcomings, the most obvious is the research capacity to create inventions in the field of pharmaceuticals. This sector as well as the production capacity of pharmaceutical enterprises in Vietnam is still limited. The data on disputes related to patent objects in the pharmaceutical sector resolved at competent authorities also shows that the legal system and enforcement agencies are not really create confidence for rights holders in protecting their legitimate rights and interests. The proposed complete solutions are put forward to supplement the gaps in IPR protection for pharmaceutical inventions and at the same time increase the community's access to medicines for health protection and care.

LIST OF SCIENTIFIC RESEARCH RELATED TO THE DOCTORAL THESIS

1. “*Protection of intellectual property rights for pharmaceutical inventions in accordance with Vietnamese law*”, Educational and social magazine, No 98 (159) 5/2019, ISSN 1859 – 3917, page 63-68, 84-85;
2. “*Balancing intellectual property rights for pharmaceutical inventions and public health issues – solutions for Vietnam*”, Jurisprudence Journal, No. 7/2019, page 92 - 104.
3. “*International and Vietnamese laws on the protection of intellectual property rights for traditional medicine*”, State and Law Journal, No. 11/2019, page 27 - 34.
4. “*Flexible provisions of the TRIPS Agreement relating to pharmaceutical patent protection and its application in Vietnam*”, Jurisprudence Journal, No. 1/2019, page 56 -71.
5. “*The protection of intellectual property rights for pharmaceutical invention in Vietnam*”, The 15th Colloquium for Teachers of Intellectual Property organized by the World Intellectual Property Organization (WIPO) and the World Trade Organization (WTO) Research paper, 6/2018.